

Differential Calculus — Chain Rule

Chapter 2: Cosine of Linear Function | Page 33

PROBLEM STATEMENT

Cosine Function & Negative Sign

Find the derivative with respect to x of the function:

$$y = \cos(5x - 1)$$

STEP 1 প্রদত্ত কোসাইন ফাংশনটি উল্লেখ করি (Given Trigonometric Function)

$$y = \cos(5x - 1)$$

STEP 2 উভয়পক্ষে x -এর সাপেক্ষে অন্তরীকরণ করি (Differentiating both sides w.r.t. x)

$$\frac{dy}{dx} = \frac{d}{dx} [\cos(5x - 1)]$$

STEP 3 কোসাইন ফাংশনের ওপর চেইন রুল প্রয়োগ করি ($\frac{d}{dx}[\cos u] = -\sin u \frac{du}{dx}$)

$$\frac{dy}{dx} = -\sin(5x - 1) \cdot \frac{d}{dx}(5x - 1)$$

STEP 4 ভেতরের রৈখিক কোণের অন্তরক নির্ণয় করি (Differentiating Inner Linear Argument)

$$\frac{d}{dx}(5x - 1) = 5(1) - 0 = 5$$

STEP 5 মান বসিয়ে ধ্রুবক সহগকে চিহ্নের সাথে গুণ করে চূড়ান্ত অন্তরজ পাই (Final Evaluation)

$$\frac{dy}{dx} = -\sin(5x - 1) \times 5 = -5 \sin(5x - 1)$$

FINAL ANSWER
$$\frac{dy}{dx} = -5 \sin(5x - 1)$$

Rigorous Insights & Mark Scheme

AMIE Section-A Standard Evaluation Protocol | Page 33, HW 05

✚ BOX 1: FORMULATION & SIGN RULES

- **Standard Formula:** $\frac{d}{dx} \cos(ax + b) = -a \sin(ax + b)$.
- **Negative Sign Rule:** All trigonometric co-functions (cos, cot, csc) always produce negative derivatives upon differentiation.
- Here, $a = 5, b = -1 \implies \frac{dy}{dx} = -5 \sin(5x - 1)$.

⚠ BOX 2: COMMON STUDENT TRAPS & PITFALLS

- **Dropping the Negative Sign:** Forgetting that $\frac{d}{dx} \cos x = -\sin x$ and erroneously writing $+5 \sin(5x - 1)$.
- **Dropping the Chain Factor 5:** Writing $-\sin(5x - 1)$ by forgetting to multiply by the inner derivative 5.

🌐 BOX 3: REAL DOMAIN & HARMONIC PROPERTIES

- **Domain & Range:** Domain is $\mathbb{R} = (-\infty, \infty)$. Amplitude of $\frac{dy}{dx}$ is 5, so range of derivative is $[-5, 5]$.
- **Fundamental Period:** $T = \frac{2\pi}{5}$.
- **Higher Derivatives:** The n -th derivative is:

$$\frac{d^n y}{dx^n} = 5^n \cos\left(5x - 1 + \frac{n\pi}{2}\right)$$

💡 BOX 4: LEIBNIZ SUBSTITUTION (ALTERNATIVE VERIFICATION)

- Let $u = 5x - 1 \implies y = \cos u$.
- $\frac{dy}{du} = -\sin u$.
- $\frac{du}{dx} = 5$.
- By Chain Rule:

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = (-\sin u) \cdot 5 = -5 \sin(5x - 1)$$

📊 BOX 5: AMIE EXAMINATION MARKS DISTRIBUTION (TOTAL: 3 / 4 MARKS)

- **1.0 Mark:** Correct identification of cosine derivative with negative sign $-\sin u$.
- **1.0 Mark:** Accurate calculation of inner derivative $\frac{d}{dx}(5x - 1) = 5$.
- **1.0 Mark:** Final result correctly framed as $-5 \sin(5x - 1)$.